

MORPHOLOGY OF MANDIBULAR PREMOLARS

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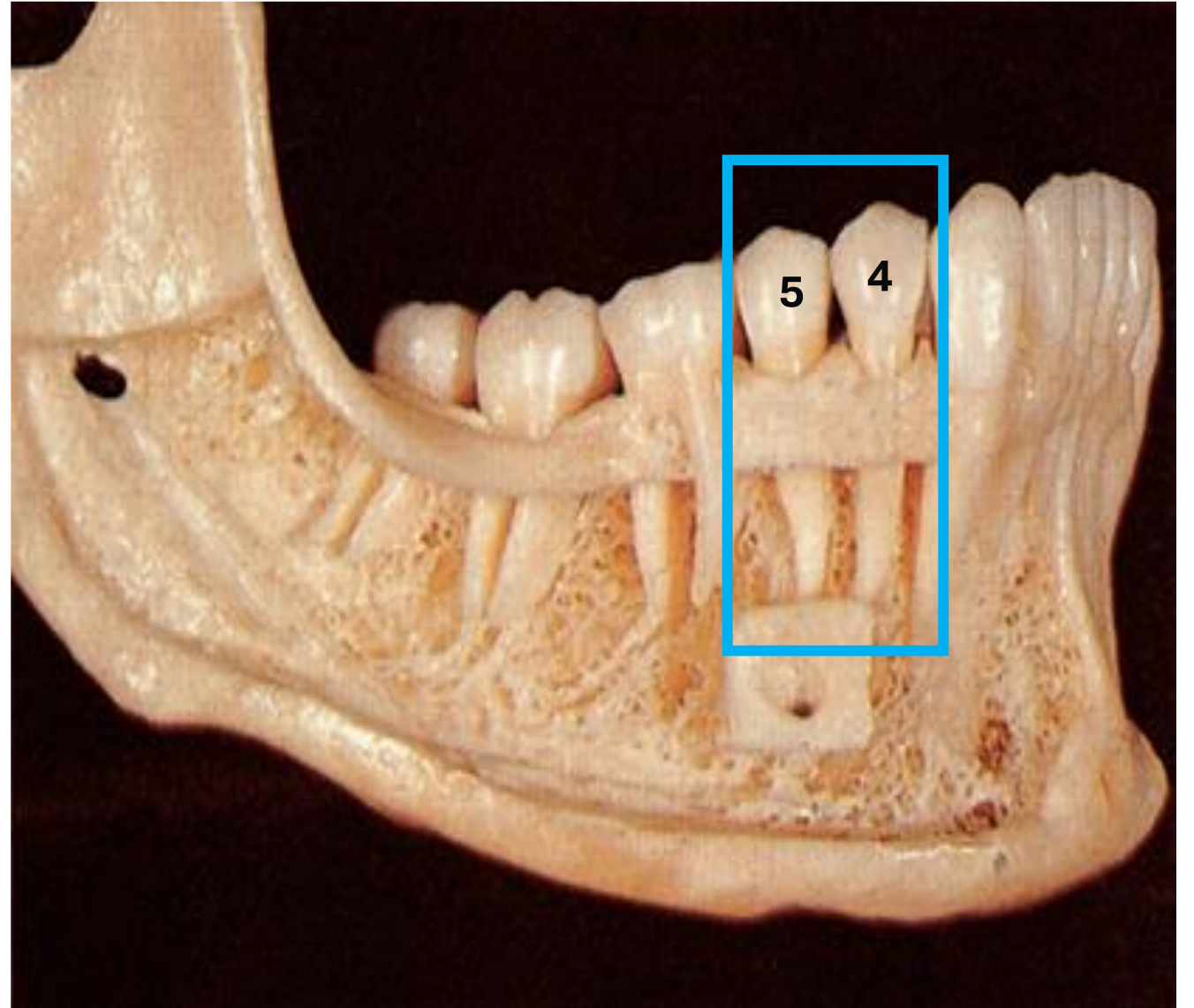
DEPT OF ORAL BIOLOGY

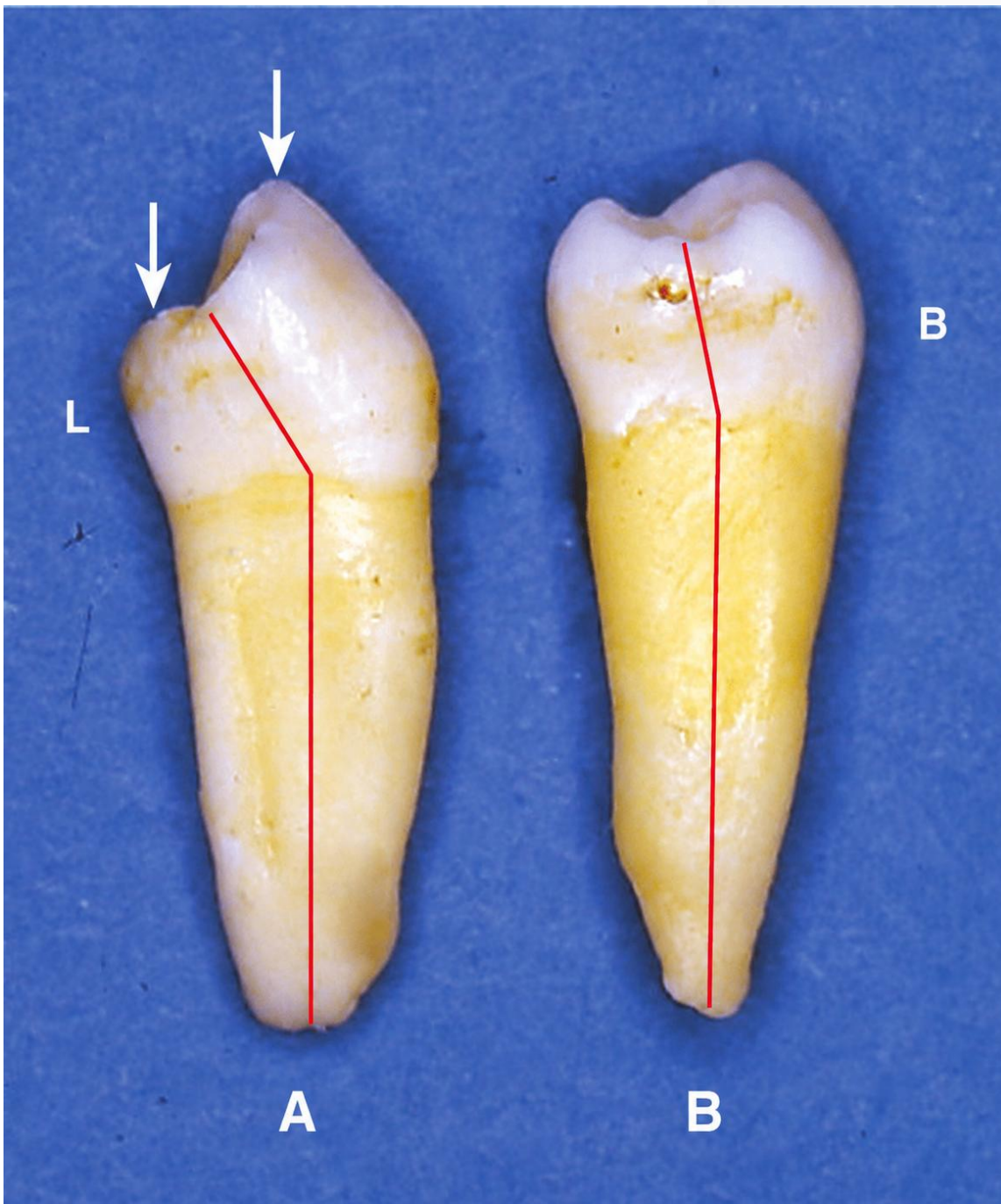
LEARNING OBJECTIVES

- To identify the positive and negative anatomical landmarks of mandibular premolars.
- To distinguish mandibular first premolar from the mandibular second premolar.
- To distinguish maxillary premolars from the mandibular premolars.
- To be to draw the graph outlines.

INTRODUCTION

The mandibular premolars number four, immediately posterior to the mandibular canines and anterior to the molars and considered as a transitional tooth.





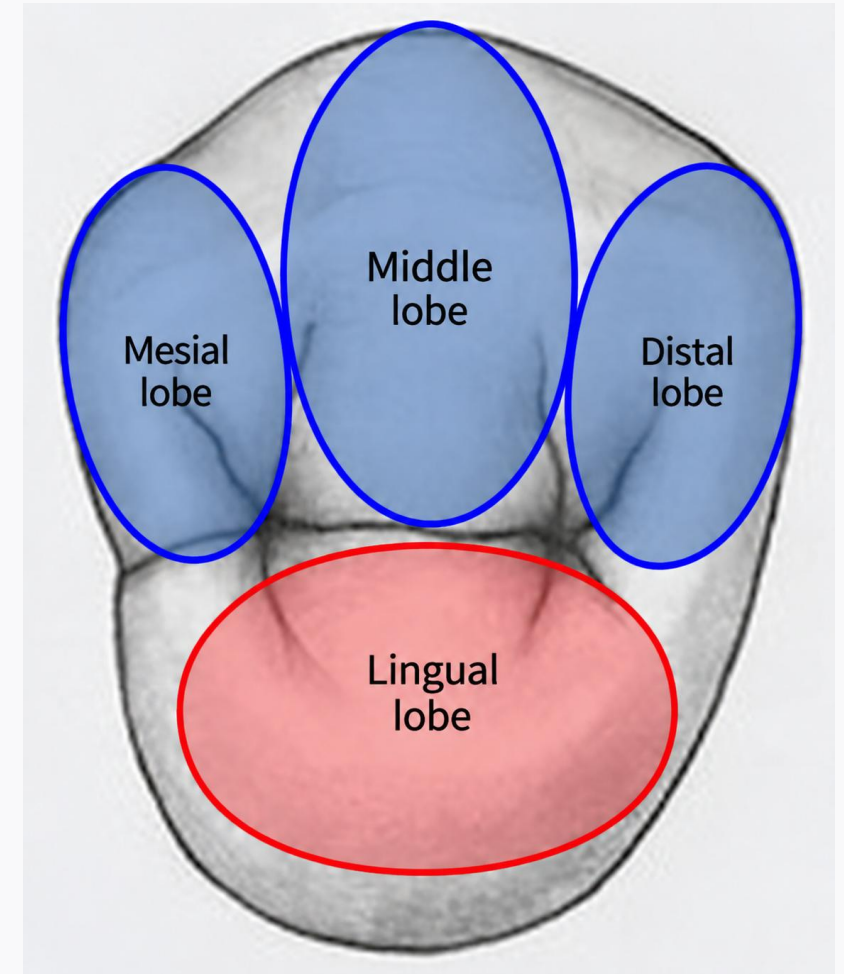
KEY CONSIDERATIONS

- A single, well-developed facial cusp, the lingual cusp or cusps may vary in size and development and are not as well developed as the facial cusp.
- From the proximal aspect, the occlusal surface slopes lingually because of the difference in cusp height.

A: mandibular first premolar

B: mandibular second premolar

	Facial Lobes	Lingual Lobes	Total Lobes
First Premolar	3	1	4
Second Premolar (two-cusp type)	3	1	4
Second Premolar (three-cusp type)	3	2	5

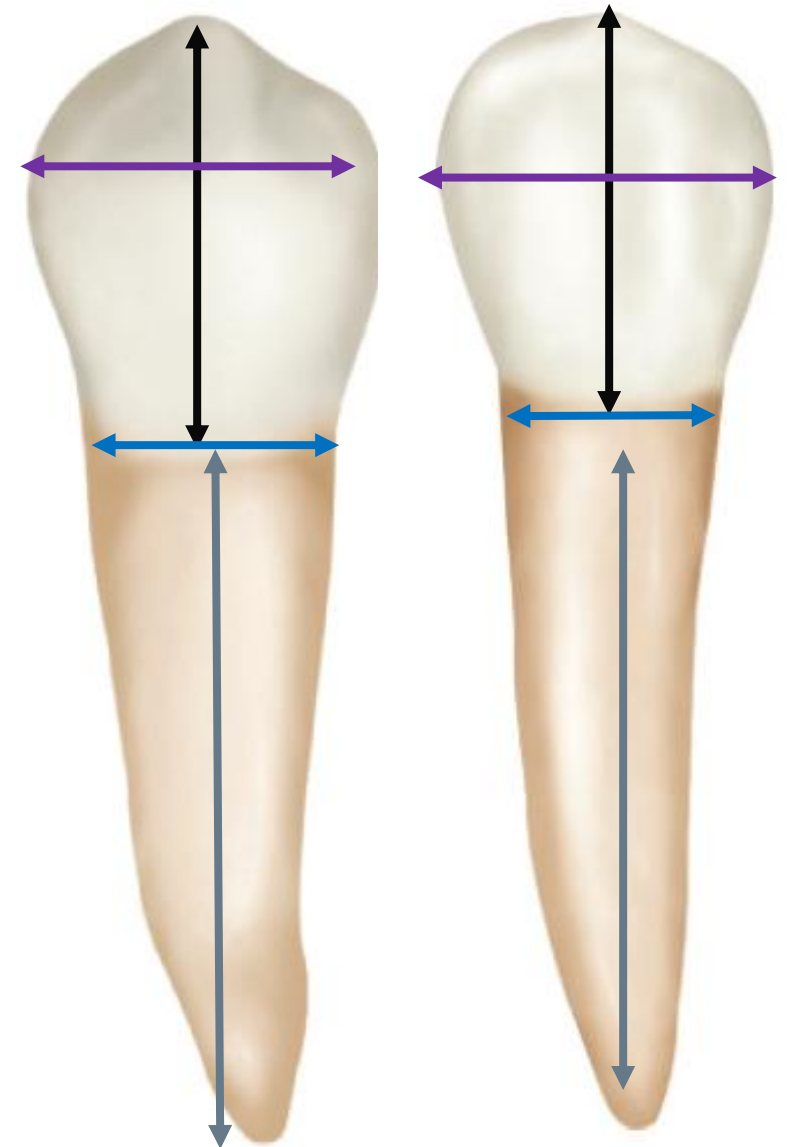


CHRONOLOGY OF MANDIBULAR PREMOLARS

	Initiation of Calcification	Crown Completion	Eruption	Root Completion
First Premolar	1¾-2 year	5-6 year	10-12 year	12-13 year
Second Premolar	2¼-2½ year	6-7 year	11-12 year	13-14 year

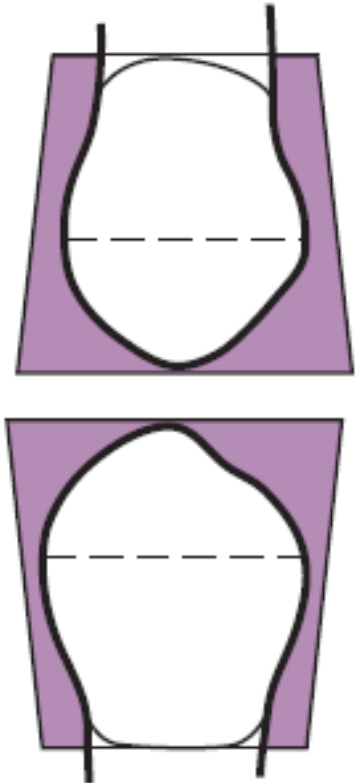
DIMENSIONS OF MANDIBULAR PREMOLARS

DIMENSIONS	FIRST PREMOLAR	SECOND PREMOLAR
Cervico-incisal length of crown	8.5 mm	8 mm
Length of root	14 mm	14.5 mm
Mesiodistal diameter of Crown	7 mm	7.2 mm
Labio-lingual diameter of crown	7.5 mm	8 mm

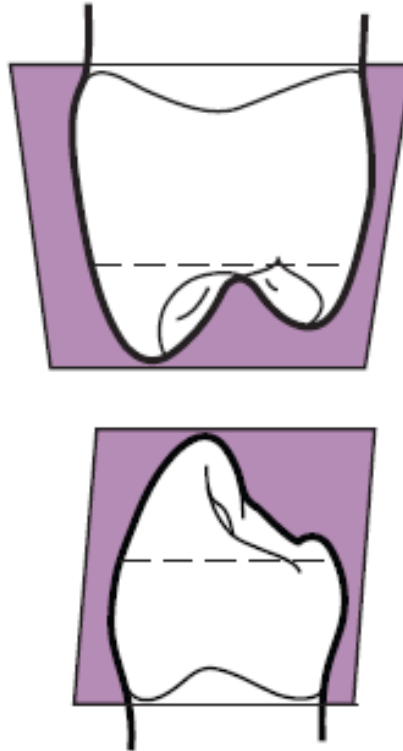


GEOMETRIC OUTLINES OF PREMOLARS

A



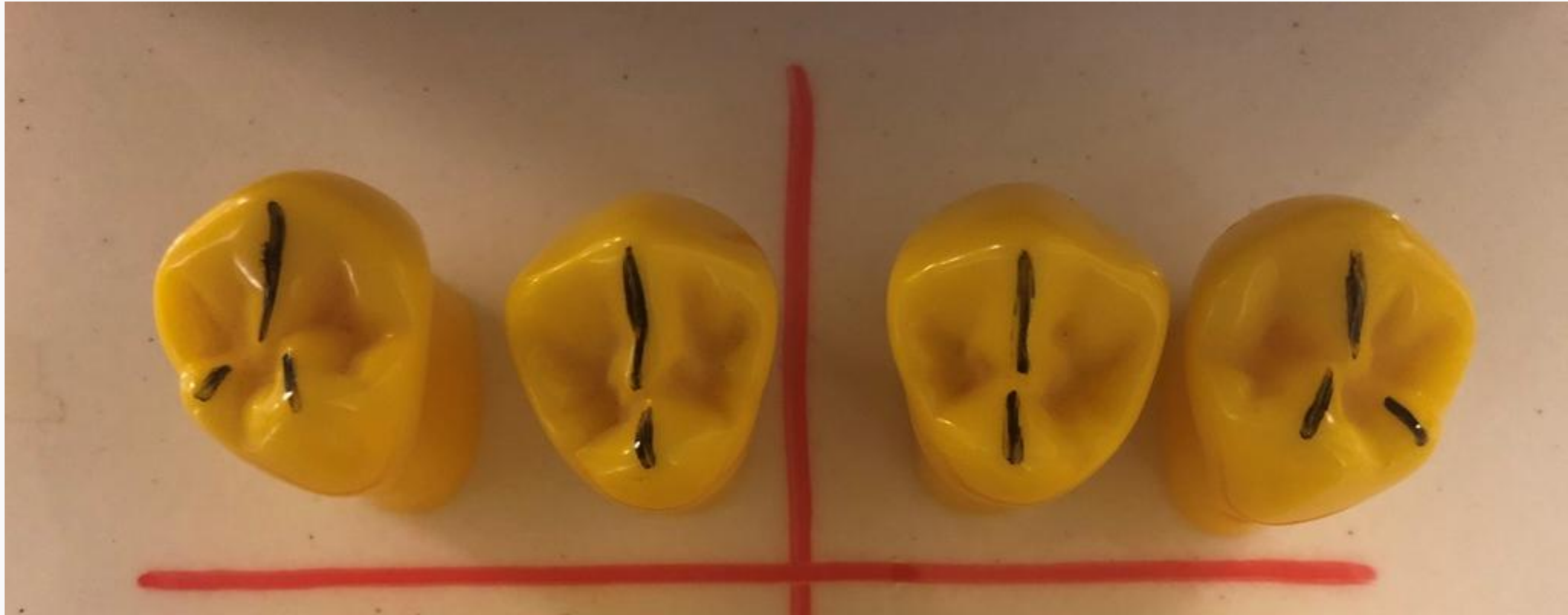
B



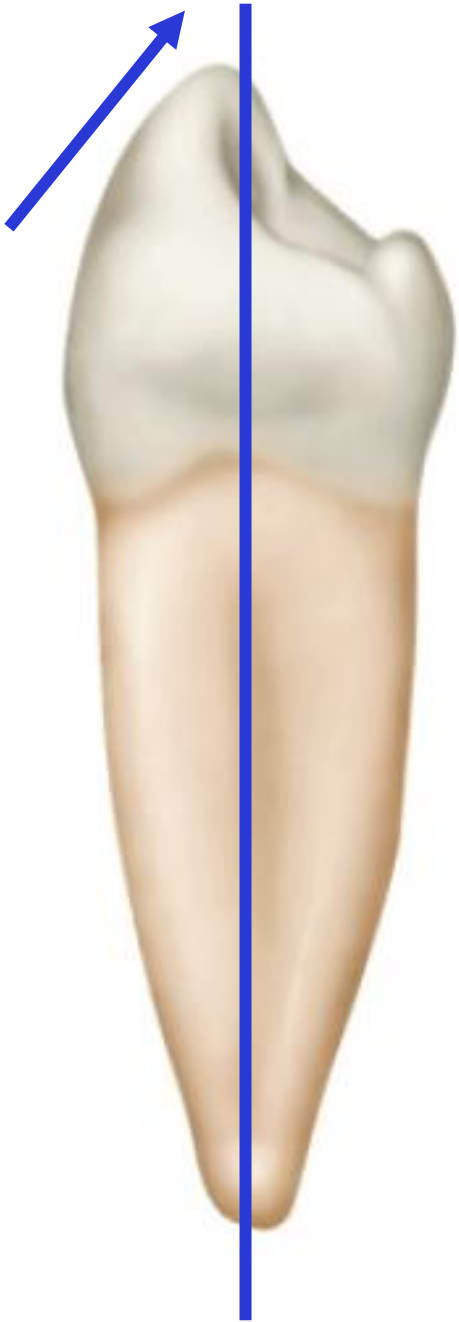
A: Buccal/Lingual: Trapezoid

B: Mesial/Distal: Rhomboid

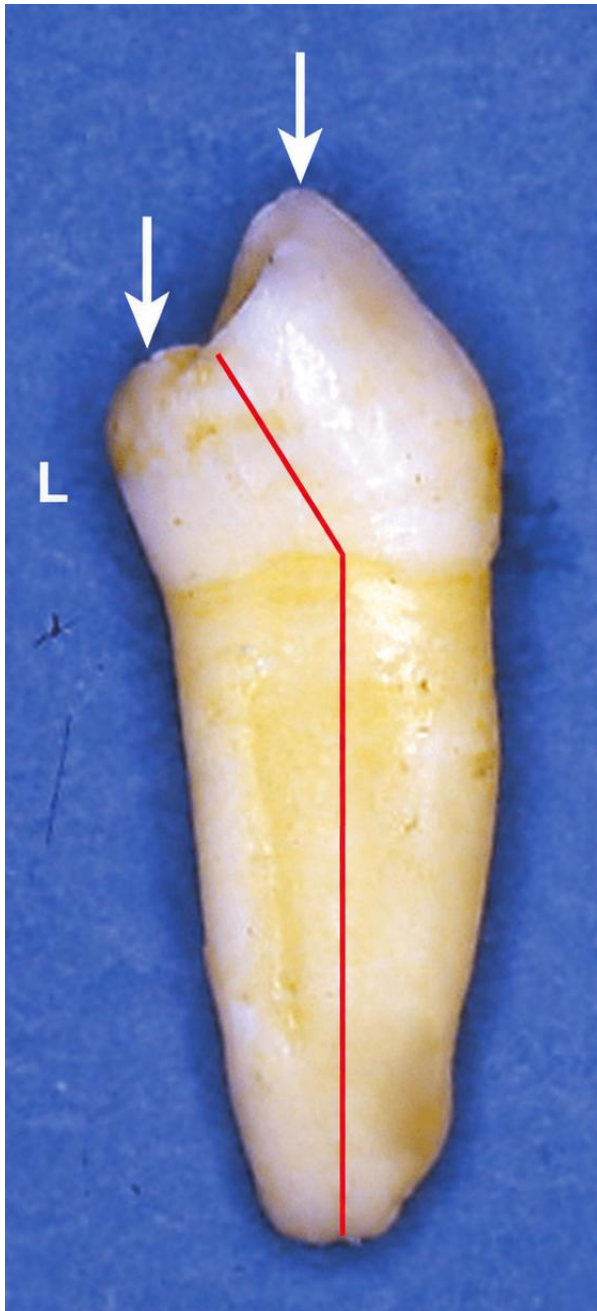
MORPHOLOGY OF MANDIBULAR FIRST PREMOLAR



First premolar is always smaller than the second premolar



The buccal cusp is long and sharp with lingual inclination.



It has one well developed buccal cusp and one non-functioning lingual cusp.

BUCCAL ASPECT

- The middle buccal lobe is well developed, resulting in a pointed buccal cusp.
- The crown is roughly trapezoidal.
- From this aspect, the tooth has near-symmetrical mesial & distal profiles.
- The mesiobuccal cusp ridge is shorter than the distobuccal cusp ridge.



BUCCAL ASPECT

- The contact areas are almost at the same level mesially and distally.
- The mesial outline of the crown is straight or slightly concave above the cervical line to a point where it joins the curvature of the mesial contact area.
- The tip of the buccal cusp is pointed and located mesial to the center of the crown.



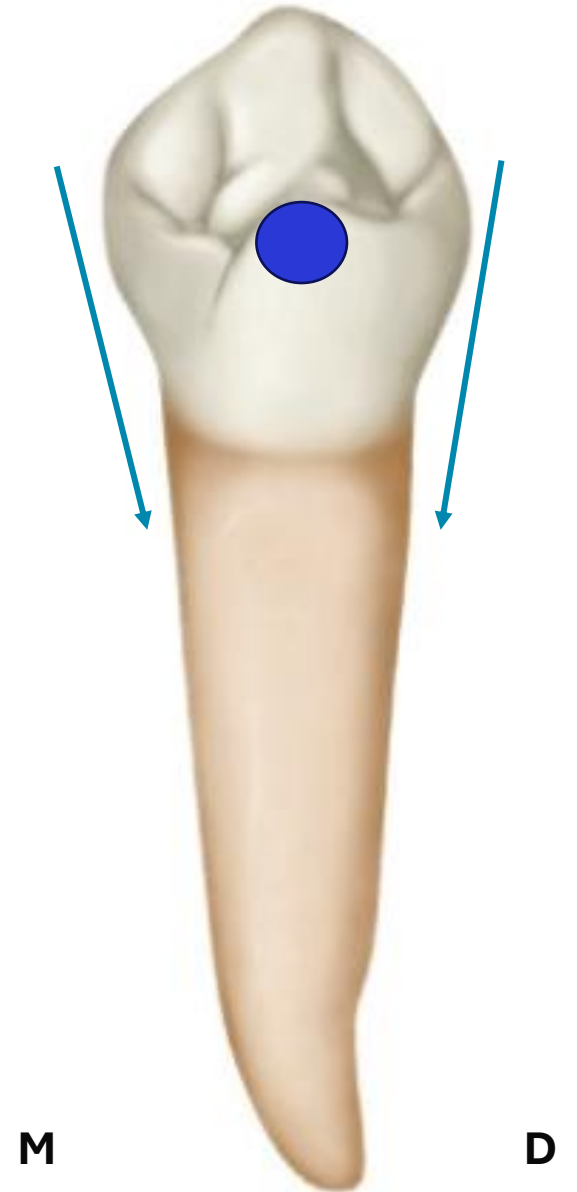
BUCCAL ASPECT

- The distal outline of the crown is slightly concave above the cervical line.
- The continuous ridge from the cervical margin to the cusp tip is called the buccal ridge.
- Roots are generally tapered, often with a distal curvature toward the apex.



LINGUAL ASPECT

- The lingual cusp is always small and non-functional.
- The crown and root taper markedly toward the lingual, so most of the mesial and distal surfaces are seen.
- The occlusal table is also visible from this aspect.



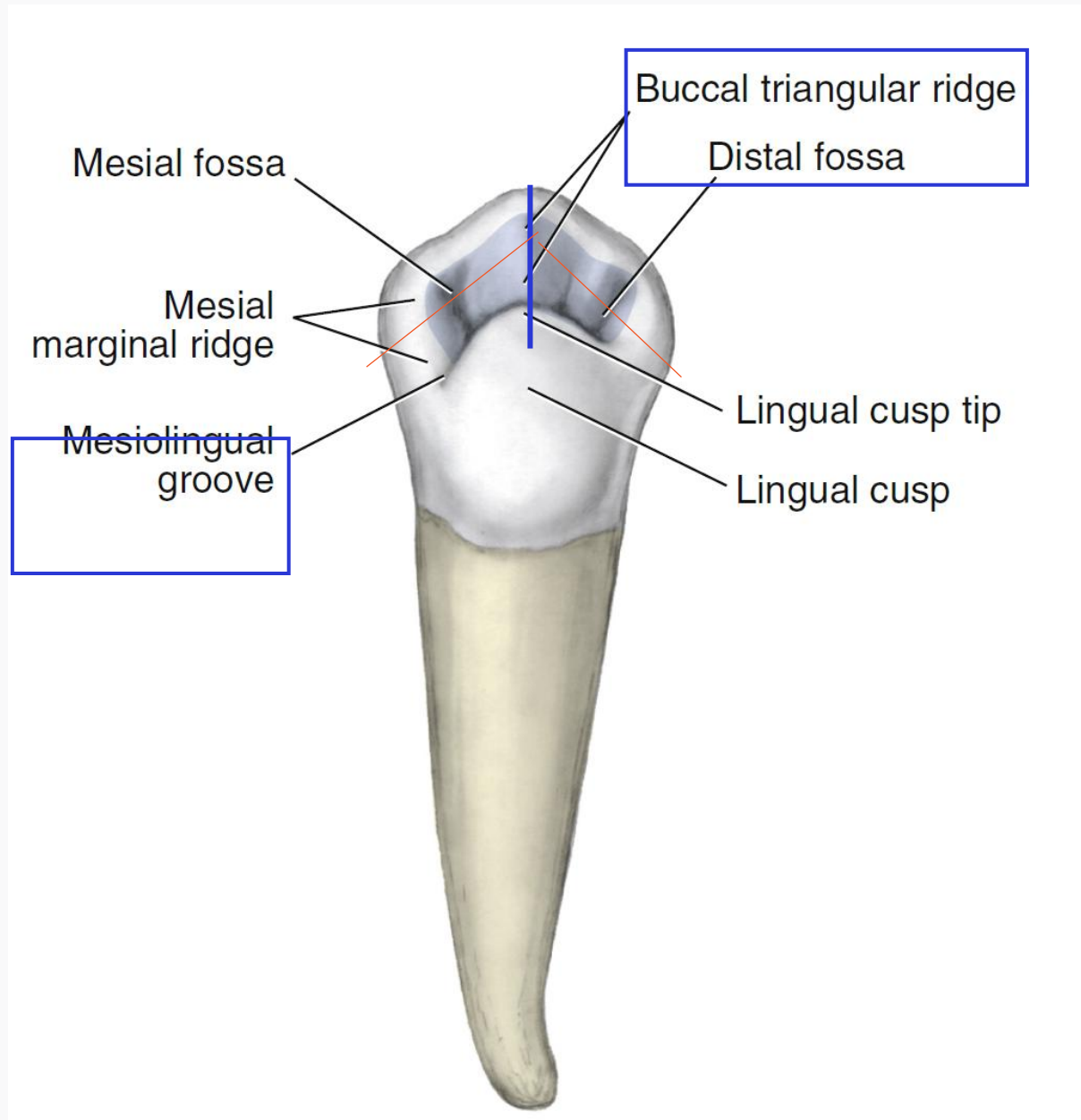
LINGUAL ASPECT

- The lingual cusp tip is in alignment with the triangular ridge.
- The root from this view is much narrower and has a pointed apex.
- There is a **mesiolingual developmental groove**.



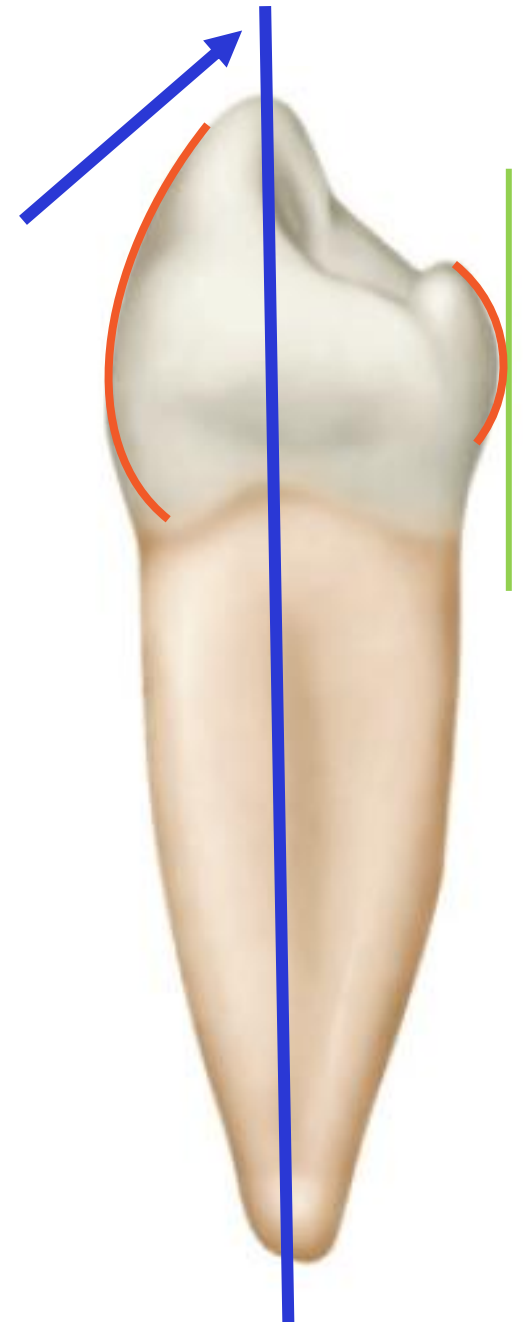


A characteristic of the lingual surface is the **mesiolingual developmental groove**. This groove acts as a line of demarcation between the mesiobuccal lobe and the lingual lobe and extends into the mesial fossa of the occlusal surface.



MESIAL ASPECT

- The buccal cusp is centered over the root from the line of vision.
- The buccal outline of the crown is curved from the cervical line to the tip of the buccal cusp.
- The buccal cusp inclines lingually.
- The lingual outline of the crown is a curved outline of less convexity.
- The surface of the crown lingually presents an overhang above the root trunk.



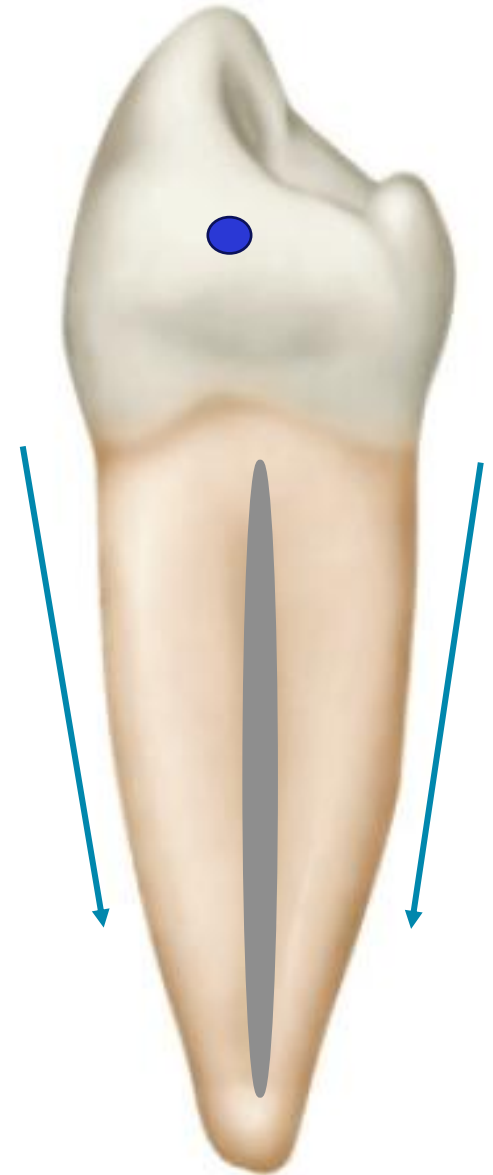
MESIAL ASPECT

- The lingual border of the mesial marginal ridge merges with the developmental depression that harbors the **mesiolingual developmental groove**.
- The cervical line on the mesial surface curves occlusally.
- The crest of the curvature is centered buccolingually.



MESIAL ASPECT

- The mesial contact area is centered in line with the buccal cusp.
- The root is tapered from the cervix, ending in a pointed apex in line with the tip of the buccal cusp.
- The root is smooth and flat from the buccal margin to the center. Then, converges sharply toward the root center displaying a **deep developmental groove**.



DISTAL ASPECT

- The distal marginal ridge is higher, being at right angles to the axis of the tooth.
- There is no groove present at the marginal ridge, and it is confluent with the cusp ridge.
- The distal contact area is broader, midway b/w cervical line and the tip of buccal cusp.
- A shallow developmental depression is centered on the root.



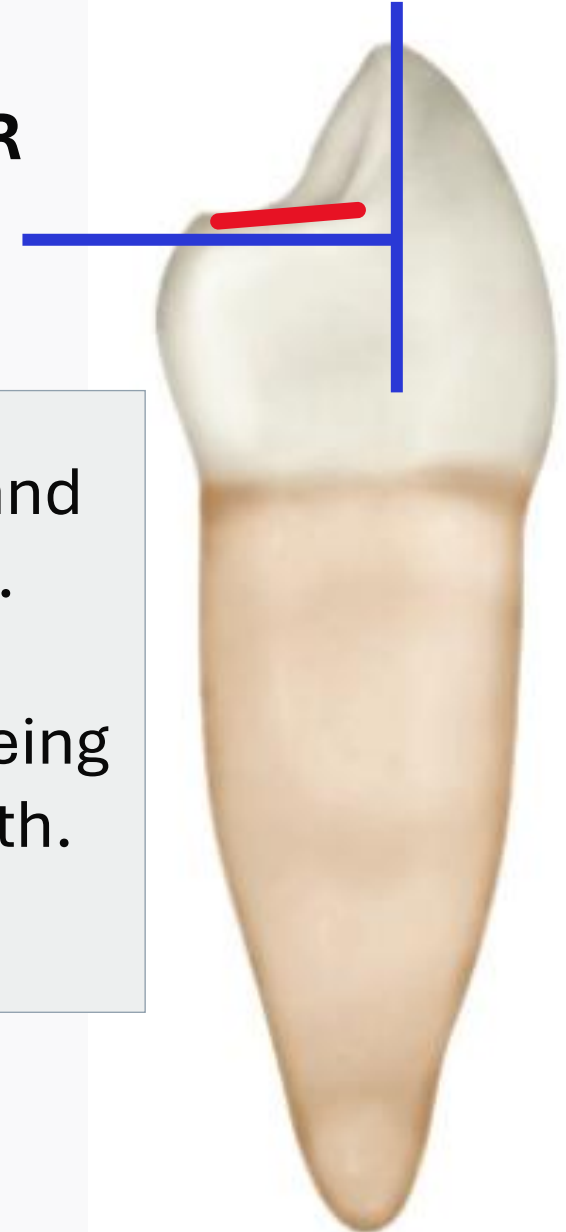
MANDIBULAR FIRST PREMOLAR MARGINAL RIDGES



M

The mesial marginal ridge is lower and parallel to buccal triangular ridge.

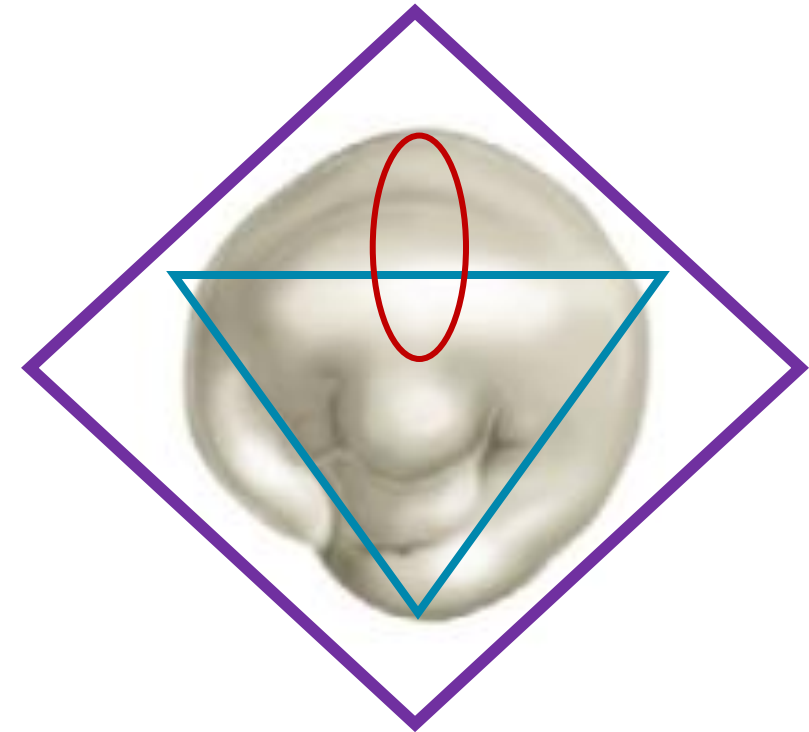
The distal marginal ridge is higher, being at right angles to the axis of the tooth.



D

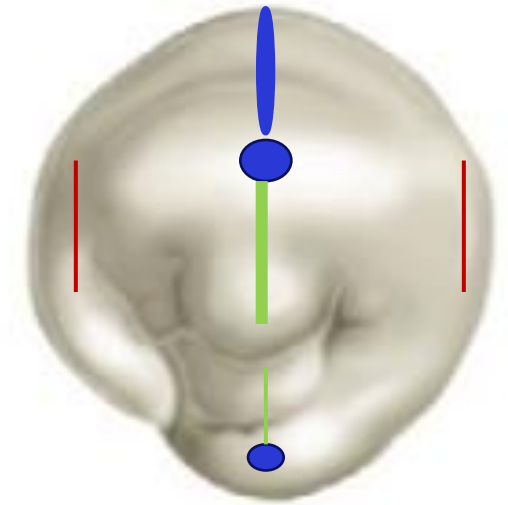
OCCLUSAL ASPECT

- The occlusal outline is diamond-shaped with a triangular occlusal table.
- The crown converges to the lingual, making the cuspal ridges, marginal ridges, and lingual lobe triangular in form.
- The middle buccal lobe makes up the major bulk of the tooth crown.



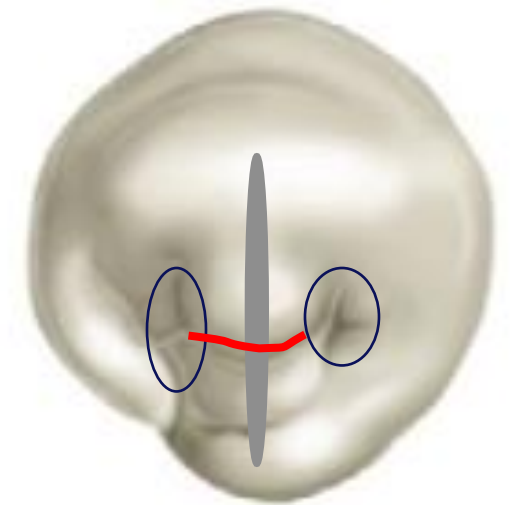
OCCLUSAL ASPECT

- The buccal ridge is prominent.
- The mesiobuccal and distobuccal line angles are prominent and rounded.
- The marginal ridges are well developed.
- The lingual cusp is small.
- A heavy buccal triangular ridge and a small lingual triangular ridge.



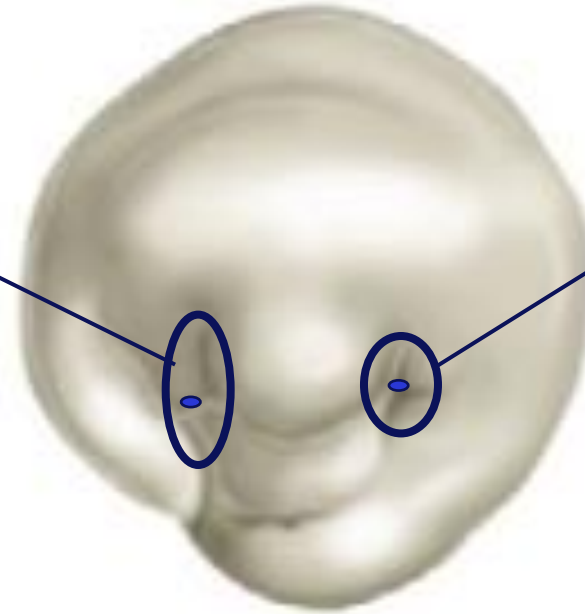
OCCLUSAL ASPECT

- There is a prominent transverse ridge from the facial cusp tip to the lingual cusp.
- Two irregular depressions: mesial and distal fossae are present.
- Because of the prominent transverse ridge, the central groove is poorly defined (in most cases).

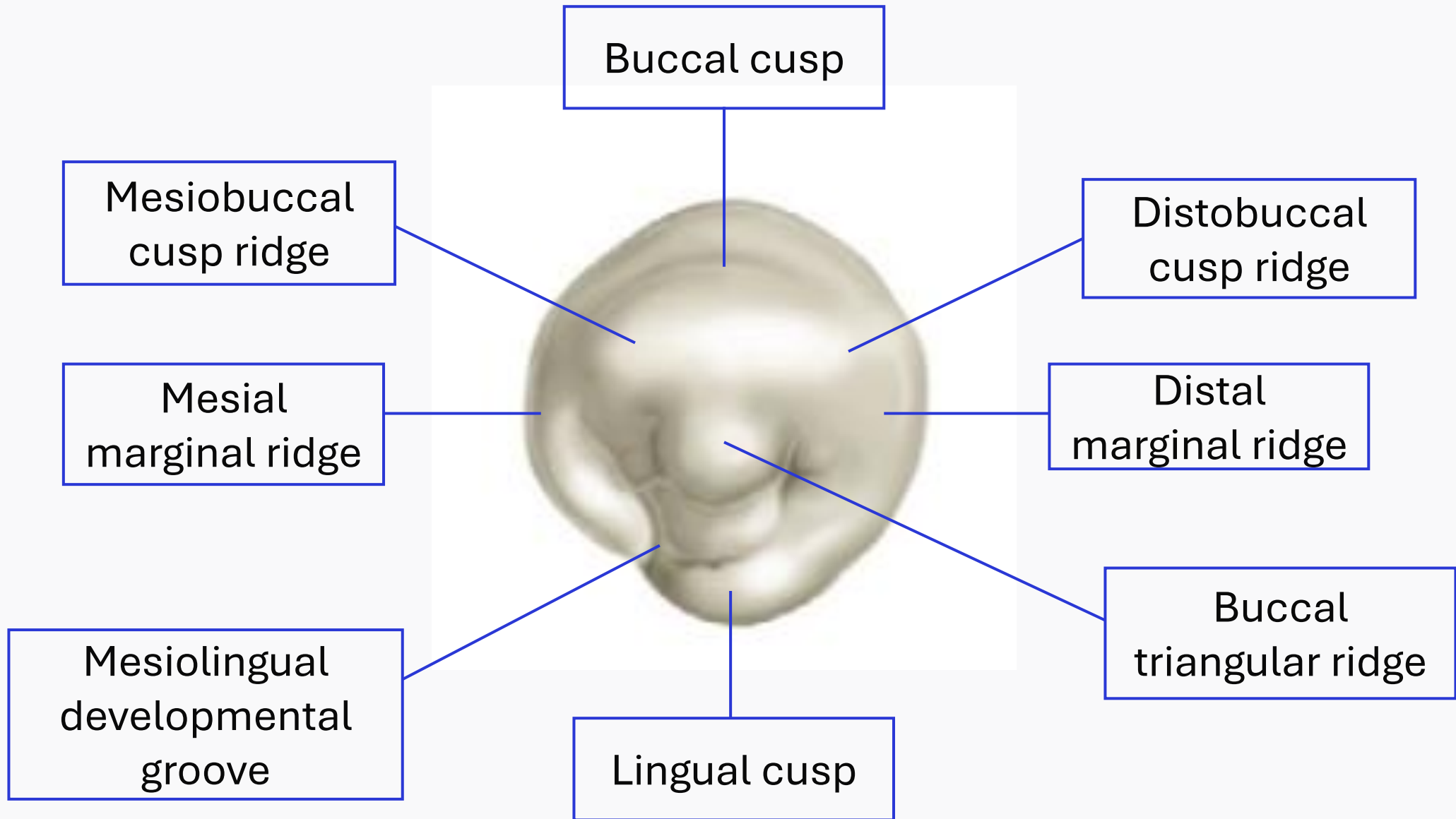


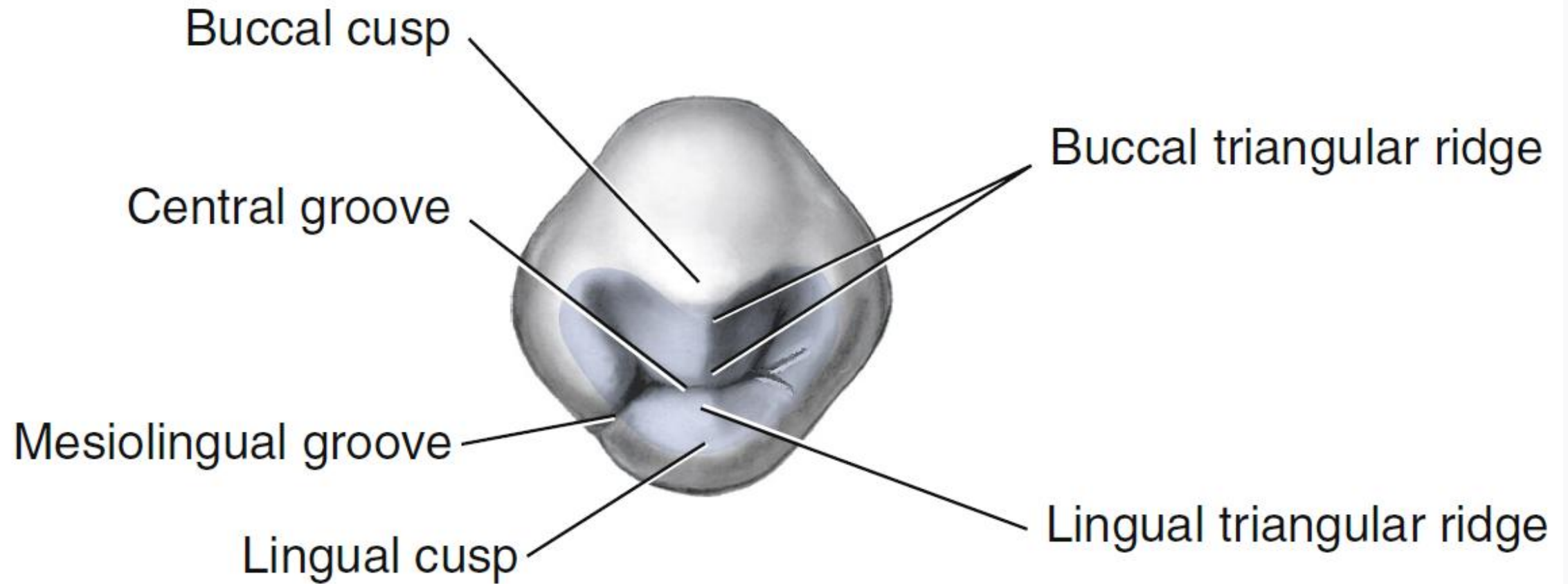
FOSSAE AND PITS

The **mesial fossa** is more linear in form, being more sulcate and containing the mesial developmental groove, which extends buccolingually.

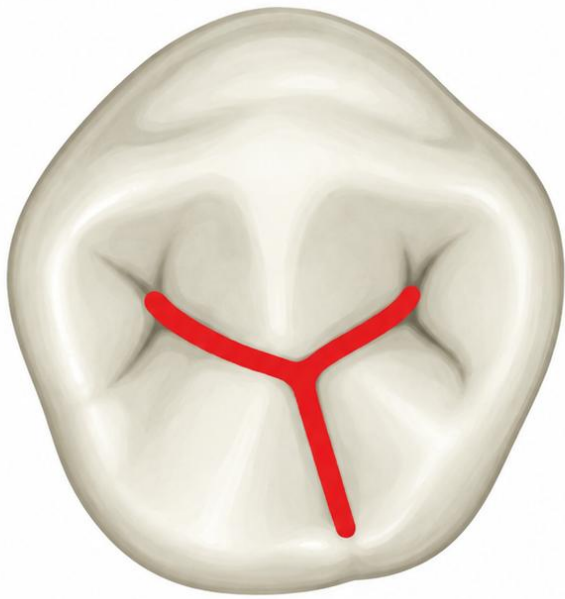


The **distal fossa** is more circular and bounded by the distobuccal cusp ridge, distal marginal ridge, buccal triangular ridge, and the distolingual cusp ridge.





MORPHOLOGY OF MANDIBULAR SECOND PREMOLAR



Three-cusp type



Two-cusp type

Three-cusp type: 3 cusps, Y-shaped groove pattern, square outline.

Two-cusp type: 2 cusps, U-shaped groove pattern, round outline.

BUCCAL ASPECT

- A shorter crown and shorter buccal cusp than the first premolar.
- The mesiobuccal is shorter than the distobuccal cusp ridge.
- The contact areas (mesial and distal) are broad and higher because of the short buccal cusp.

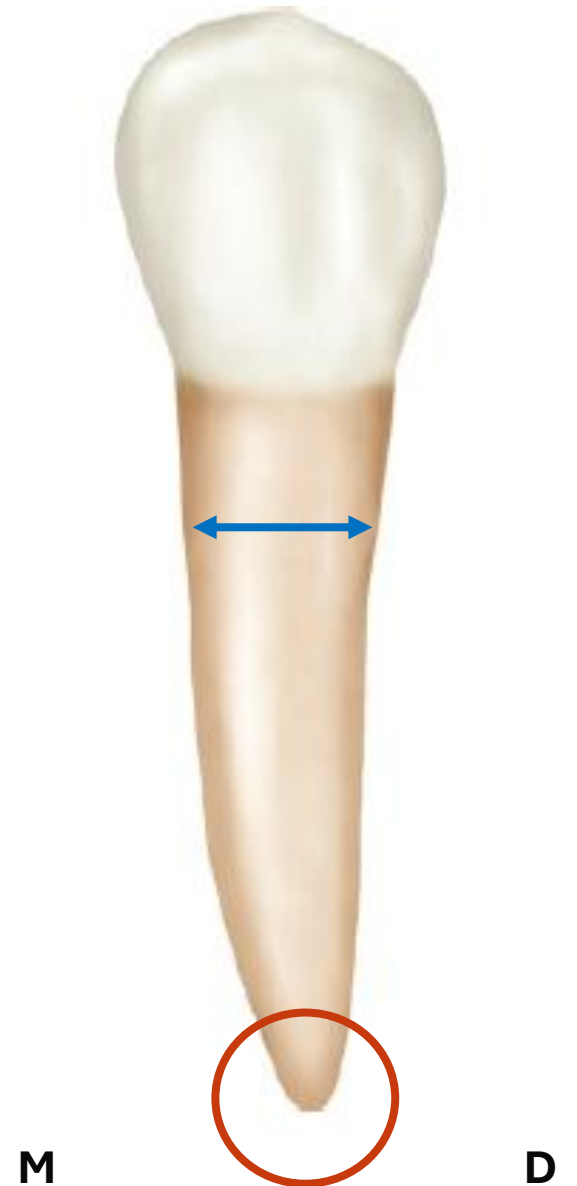


M

D

BUCCAL ASPECT

- The root is broader mesiodistally than that of the first premolar.
- The root is longer (0.5mm) than that of the first premolar, with a blunt apex.



LINGUAL ASPECT

- Smooth, spheroidal, and bulbous form.
- Two lingual cusps are present: mesiolingual (larger) and distolingual.
- A lingual developmental groove divides the two lingual cusps.
- Much less convergence toward the lingual and less view of occlusal surface is seen when compared to the first premolar.



MESIAL ASPECT

- The crown and root are wider buccolingually (0.5mm) than first premolar.
- The buccal cusp is short and is not centered over the root.
- Mesial marginal ridge is higher and at right angles to the long axis of tooth.
- The root is longer (0.5mm) and root apex is blunt.



DISTAL ASPECT

- The distal marginal ridge is lower than the mesial marginal ridge.
- The occlusal table is visible from this aspect since the DMR is lower.
- The tooth appears to be tipped distally.
- There is deep distal root depression on the second premolar.





M

MANDIBULAR SECOND PREMOLAR MARGINAL RIDGES

The mesial marginal ridge is higher
and at right angles to long axis.

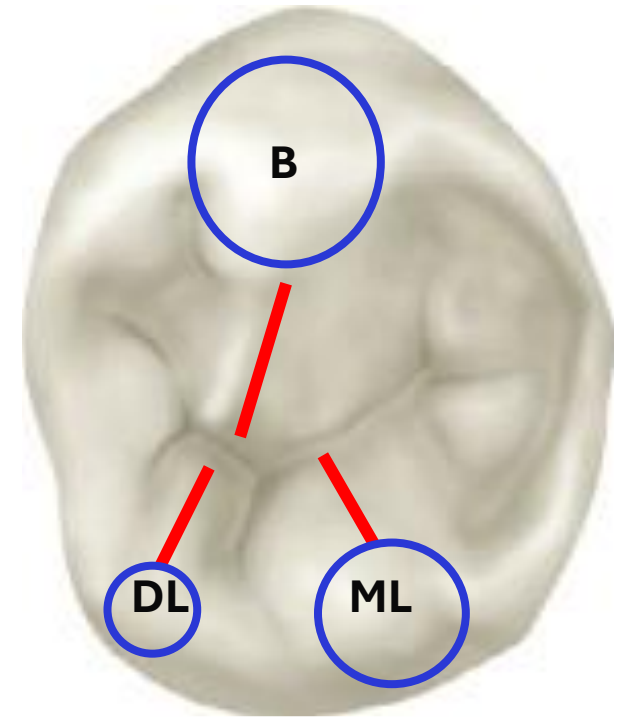
The distal marginal ridge is lower.



D

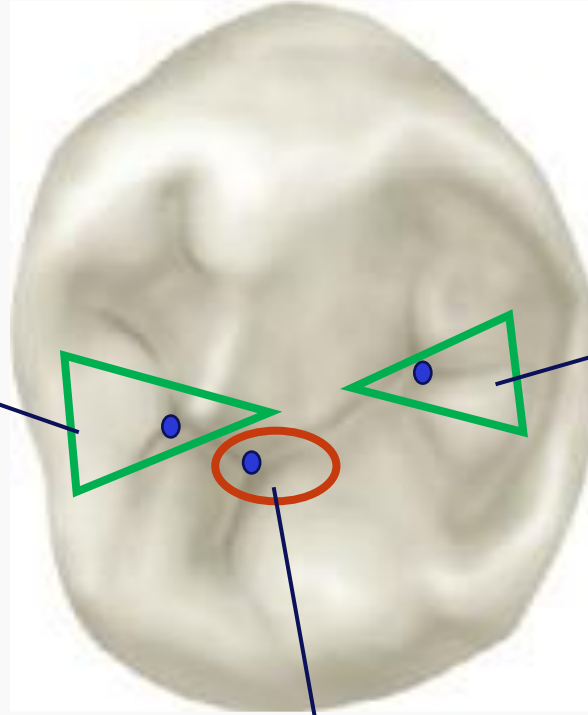
OCCLUSAL ASPECT

- The occlusal outline is square (three-cusp type).
- All the grooves converge in the central pit to form Y-shaped pattern.
- Three cusps: buccal (largest), mesio-lingual (second largest), distolingual cusp (smallest).
- Each cusp has its own triangular ridge.
- There is no transverse ridge.



FOSSAE AND PITS

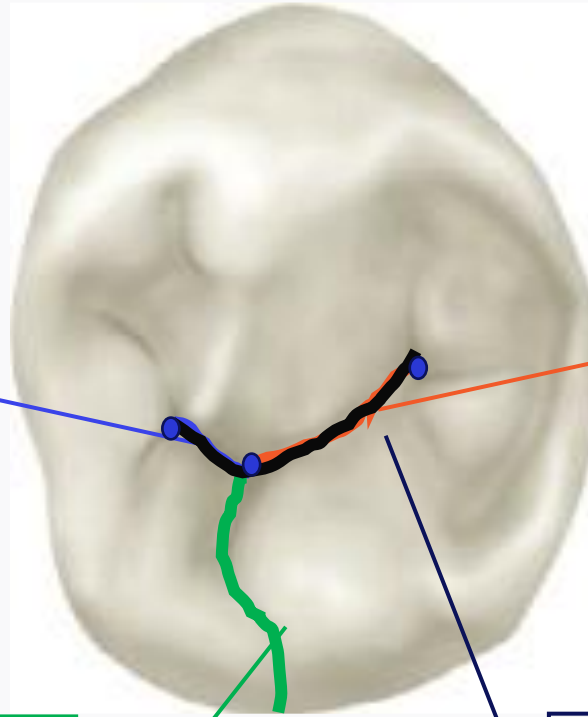
The **distal triangular fossa** is mesial to the distal marginal ridge and harbors the distal pit.



The **mesial triangular fossa** is distal to the mesial marginal ridge and harbors the mesial pit.

A well-developed small **central fossa** is present and has a central pit in the middle.

DEVELOPMENTAL GROOVES



Distal developmental groove:
From distal pit to central pit.

Mesial developmental groove:
from central pit to MTF.

Lingual developmental groove:
extends lingually b/w two lingual
cusps, distal to the center of the
crown.

Central developmental
groove: crescent-shaped
extending mesiodistally.

	MANDIBULAR FIRST PREMOLAR	MANDIBULAR SECOND PREMOLAR
DIMENSIONS	Similar dimensions expect root and buccolingual calibration.	Crown is wider buccolingually with a longer root.
BUCCAL VIEW	Buccal cusp sharp and prominent. The mesial cuspal ridge is shorter than the distal cuspal ridge. Buccal cusp tip is mesial to center. Well-developed buccal ridge. Root converges lingually, is narrower, is shorter and has a pointed apex.	Buccal cusp is short and rounded. The mesial cuspal ridge is shorter than the distal cuspal ridge. Buccal cusp tip is nearly centered. Fairly developed buccal ridge. Root is broader, longer (0.5mm) and has a blunt apex.
LINGUAL VIEW	Lingual surface is narrower. Non-functioning lingual cusp Presence of mesio-lingual developmental groove Occlusal table visible from this view Lingual convergence.	Lingual surface is broader. Well-developed lingual cusps (2). A lingual developmental groove divides the two lingual cusps. Occlusal table not visible. Slight lingual convergence.

MANDIBULAR FIRST PREMOLAR

MANDIBULAR SECOND PREMOLAR

PROXIMAL VIEWS

Much difference in buccal and lingual cusps height.
The buccal cusp is lingually inclined.
Mesio-lingual developmental groove visible from mesial side.
Mesial marginal ridge is lower and parallel to buccal triangular ridge, distal is higher
Deep developmental groove of root on mesial side.

Less difference in the heights of cusps.
The buccal cusp not inclined lingually.
No groove appears on the mesial side.
Mesial marginal ridge is higher and distal lower.
Deep developmental groove of root on distal side.

OCCLUSAL VIEW

Diamond-shaped outline.
Well-developed transverse ridge.
The mesial and distal fossae are irregular and rounded.
Poorly defined central groove.
Central fossa absent.

Square outline.
No transverse ridge.
Prominent mesial and distal triangular fossae.
Deep central groove.
Central fossa present.

THANK YOU!